



1. Description

1.1. Project

Project Name	snap
Board Name	custom
Generated with:	STM32CubeMX 6.18.1
Date	09/17/2026

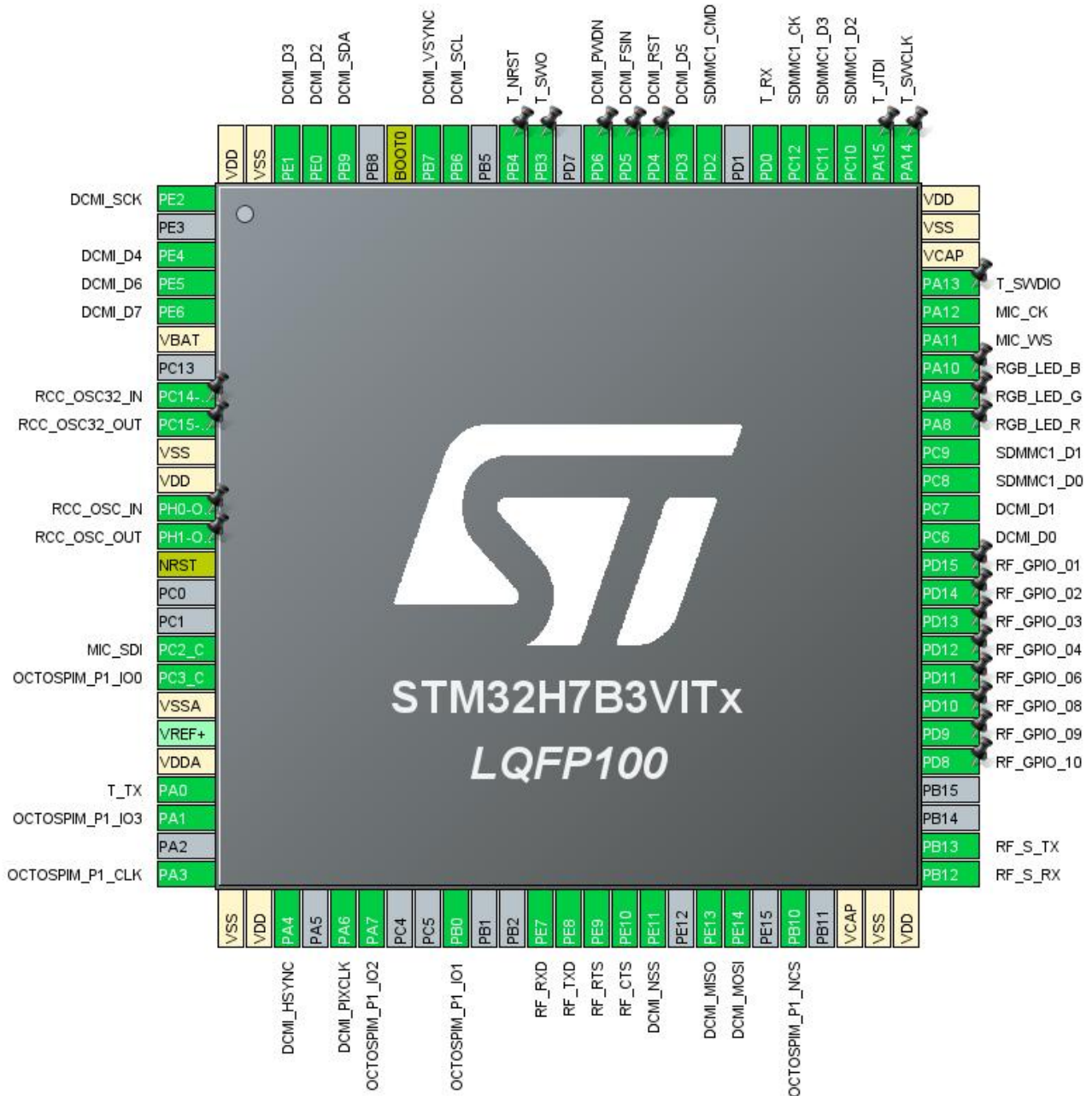
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H7A3/7B3
MCU name	STM32H7B3VITx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	Arm Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	SPI4_SCK	DCMI_SCK
3	PE4	I/O	DCMI_D4	
4	PE5	I/O	DCMI_D6	
5	PE6	I/O	DCMI_D7	
6	VBAT	Power		
8	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
9	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
14	NRST	Reset		
17	PC2_C	I/O	I2S2_SDI	MIC_SDI
18	PC3_C	I/O	OCTOSPIM_P1_IO0	
19	VSSA	Power		
21	VDDA	Power		
22	PA0	I/O	UART4_TX	T_TX
23	PA1	I/O	OCTOSPIM_P1_IO3	
25	PA3	I/O	OCTOSPIM_P1_CLK	
26	VSS	Power		
27	VDD	Power		
28	PA4	I/O	DCMI_HSYNC	
30	PA6	I/O	DCMI_PIXCLK	
31	PA7	I/O	OCTOSPIM_P1_IO2	
34	PB0	I/O	OCTOSPIM_P1_IO1	
37	PE7	I/O	UART7_RX	RF_RXD
38	PE8	I/O	UART7_TX	RF_TXD
39	PE9	I/O	UART7_RTS	RF_RTS
40	PE10	I/O	UART7_CTS	RF_CTS
41	PE11	I/O	SPI4_NSS	DCMI_NSS
43	PE13	I/O	SPI4_MISO	DCMI_MISO
44	PE14	I/O	SPI4_MOSI	DCMI_MOSI
46	PB10	I/O	OCTOSPIM_P1_NCS	
48	VCAP	Power		
49	VSS	Power		
50	VDD	Power		

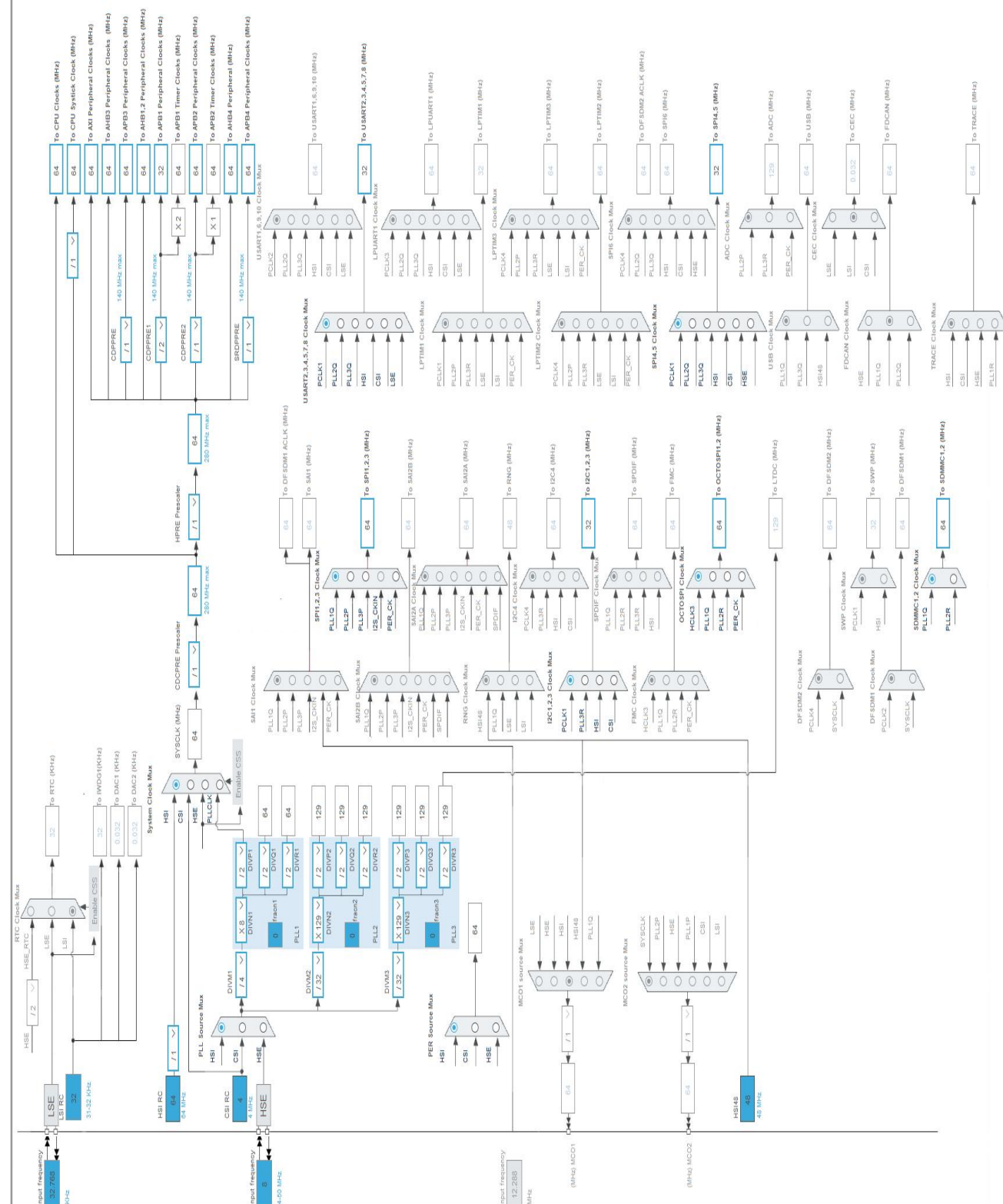
Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
51	PB12	I/O	UART5_RX	RF_S_RX
52	PB13	I/O	UART5_TX	RF_S_TX
55	PD8 *	I/O	GPIO_Output	RF_GPIO_10
56	PD9 *	I/O	GPIO_Output	RF_GPIO_09
57	PD10 *	I/O	GPIO_Output	RF_GPIO_08
58	PD11 *	I/O	GPIO_Output	RF_GPIO_06
59	PD12 *	I/O	GPIO_Output	RF_GPIO_04
60	PD13 *	I/O	GPIO_Output	RF_GPIO_03
61	PD14 *	I/O	GPIO_Input	RF_GPIO_02
62	PD15 *	I/O	GPIO_Output	RF_GPIO_01
63	PC6	I/O	DCMI_D0	
64	PC7	I/O	DCMI_D1	
65	PC8	I/O	SDMMC1_D0	
66	PC9	I/O	SDMMC1_D1	
67	PA8 *	I/O	GPIO_Output	RGB_LED_R
68	PA9 *	I/O	GPIO_Output	RGB_LED_G
69	PA10 *	I/O	GPIO_Output	RGB_LED_B
70	PA11	I/O	I2S2_WS	MIC_WS
71	PA12	I/O	I2S2_CK	MIC_CK
72	PA13	I/O	DEBUG_JTMS-SWDIO	T_SWDIO
73	VCAP	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14	I/O	DEBUG_JTCK-SWCLK	T_SWCLK
77	PA15	I/O	DEBUG_JTDI	T_JTDI
78	PC10	I/O	SDMMC1_D2	
79	PC11	I/O	SDMMC1_D3	
80	PC12	I/O	SDMMC1_CK	
81	PD0	I/O	UART4_RX	T_RX
83	PD2	I/O	SDMMC1_CMD	
84	PD3	I/O	DCMI_D5	
85	PD4 *	I/O	GPIO_Output	DCMI_RST
86	PD5 *	I/O	GPIO_Output	DCMI_FSN
87	PD6 *	I/O	GPIO_Output	DCMI_PWDN
89	PB3	I/O	DEBUG_JTDO-SWO	T_SWO
90	PB4	I/O	DEBUG_NJTRST	T_NRST
92	PB6	I/O	I2C1_SCL	DCMI_SCL
93	PB7	I/O	DCMI_VSYNC	
94	BOOT0	Boot		

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
96	PB9	I/O	I2C1_SDA	DCMI_SDA
97	PE0	I/O	DCMI_D2	
98	PE1	I/O	DCMI_D3	
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32H7B3VITx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H7A3/7B3
MCU	STM32H7B3VITx
Datasheet	DS13139 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

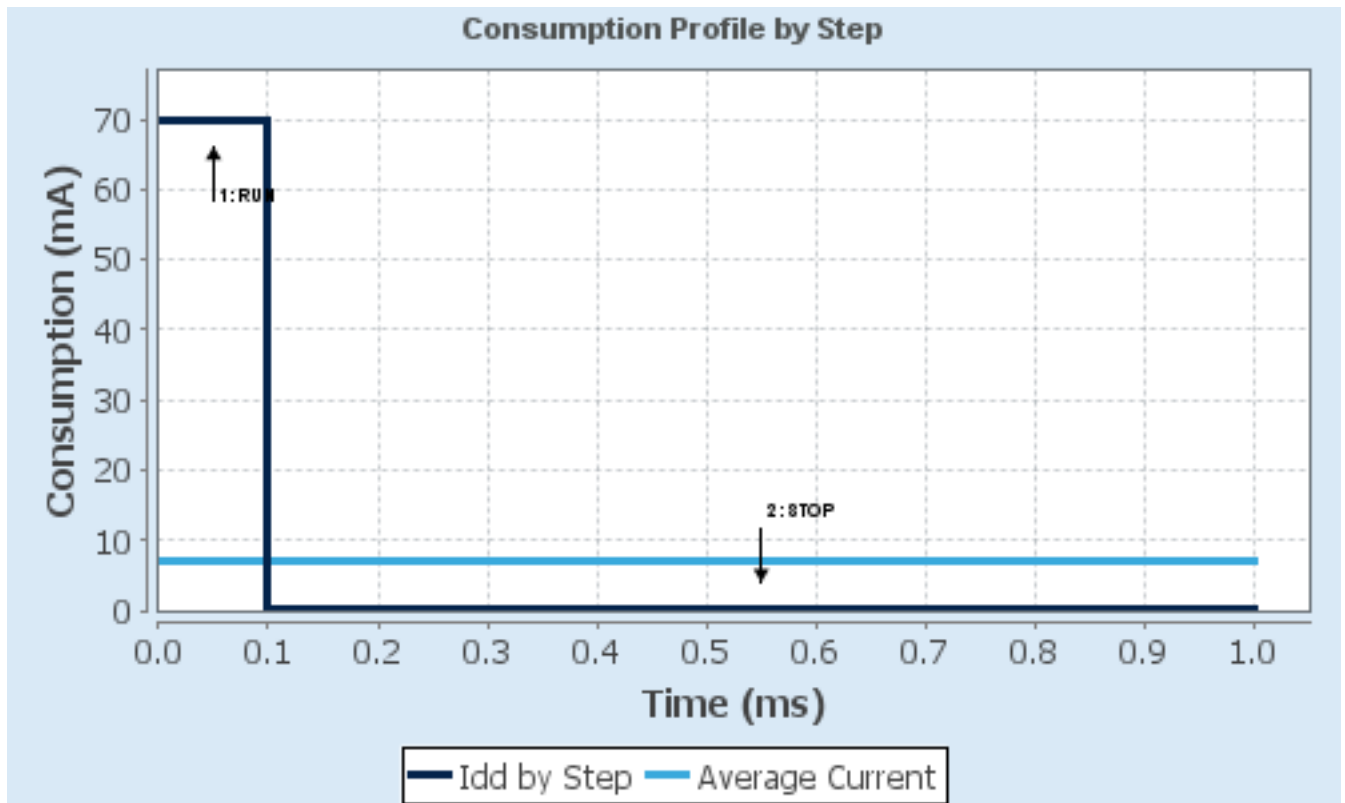
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0	SVOS5
SRDomain	DRUN	DSTOP
n/a	SRDRUN	SRDSTOP
Fetch Type	ITCM/DTCM/Cache	NA
CPU Frequency	280 MHz	64 MHz
Clock Configuration	HSE PLL	HSI Flash-ON
Clock Source Frequency	16 MHz	64 MHz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	69.92 mA	263.82 μ A
Duration	0.1 ms	0.9 ms
DMIPS	599.0	0.0
Ta Max	115.56	124.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	7.23 mA
Battery Life	19 days, 14 hours	Average DMIPS	599.2 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	snap
Project Folder	\\wsl.localhost\Ubuntu\home\rdrsstreetlight\snap
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DCMI_Init	DCMI
4	MX_UART5_Init	UART5
5	MX_UART7_Init	UART7
6	MX_OCTOSPI1_Init	OCTOSPI1
7	MX_JPEG_Init	JPEG
8	MX_UART4_Init	UART4
9	MX_I2C1_Init	I2C1
10	MX_I2S2_Init	I2S2
11	MX_SDMMC1_SD_Init	SDMMC1

Rank	Function Name	Peripheral Instance Name
12	MX_SPI4_Init	SPI4

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode **Enabled ***

Cortex Interface Settings:

CPU ICache Disabled

CPU DCache Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region Enabled

MPU Region Base Address **0x0 ***

MPU Region Size 4GB

MPU SubRegion Disable **0x87 ***

MPU TEX field level level 0

MPU Access Permission ALL ACCESS NOT PERMITTED

MPU Instruction Access DISABLE

MPU Shareability Permission ENABLE

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.2. DCMI

DCMI: Slave 8 bits External Synchro

3.2.1. Parameter Settings:

Mode Config:

Pixel clock polarity	Active on Falling edge
Vertical synchronization polarity	Active Low
Horizontal synchronization polarity	Active Low
Frequency of frame capture	All frames are captured
JPEG mode	Enabled *

Interface Capture Config:

Byte Select Mode	Interface captures all received bytes
Line Select Mode	Interface captures all received lines

3.3. DEBUG

Debug: JTAG (5 pins)

3.4. I2C1

I2C: I2C

3.4.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00707CBB *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.5. I2S2

Mode: Half-Duplex Master Receive

3.5.1. Parameter Settings:

Generic Parameters:

Communication Standard	I2S Philips
Data and Frame Format	16 Bits Data on 16 Bits Frame
Selected Audio Frequency	8 KHz
Real Audio Frequency	8.0 KHz *
Error between Selected and Real	0.0 % *

Clock Parameters:

Clock Source	PLL I2SR Clock
Clock Polarity	Low

Advanced Parameters:

First Bit	Firstbit Msb
Ws Inversion	Ws Inversion Disable
Data24 Bit Alignment	Data 24 Bit Alignment Right
Master Keep Io State	Master Keep Io State Disable

3.6. JPEG

mode: Activated

3.6.1. Parameter Settings:

Version:

JPEG version jpeg1_v1_0

JPEG Software options:

ENCODE	Enabled
DECODE	Enabled
RGB_FORMAT	JPEG_ARGB8888
JPEG_SWAP_RB	0

3.7. MEMORYMAP

mode: Activated

3.8. OCTOSPI1

Mode: Quad SPI

mode: Clock

mode: Chip Select Port1 NCS

Data [3:0]: Port1 IO[3:0]

3.8.1. Parameter Settings:

Generic:

Fifo Threshold	1
Dual Quad mode	Disable
Memory Type	Macronix *
Device Size	32
Chip Select High Time	1
Free Running Clock	Disable
Clock Mode	Low
Wrap Size	Not Supported
Clock Prescaler	1
Sample Shifting	None
Delay Hold Quarter Cycle	Disable
Chip Select Boundary	0
Delay Block	Disable
Maximum Transfer	0
Refresh Rate	0

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

CRS SYNC: CRS SYNC Source LSE

3.9.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 3

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability
CSI Calibration Value	16
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	2 WS (3 CPU cycle)

PLL range Parameters:

PLL1 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range

CRS Parameters:

CRS Synchro Divider	1
CRS Synchro Polarity	Active on rising edge
CRS Synchro Reload Value Type	Automatic
CRS Synchro frequency (Hz)	32768
Error limit Value	34
HSI48 Calibration Value	32

3.10. SDMMC1

Mode: SD 4 bits Wide bus

3.10.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
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SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.11. SPI4

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Input Signal

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	16.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Input Hardware
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.12. SYS

Timebase Source: SysTick

3.13. UART4

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. UART5

Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. UART7

Mode: Asynchronous

Hardware Flow Control (RS232): CTS/RTS

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable

DMA on RX Error	Enable
MSB First	Disable

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DCMI	PE4	DCMI_D4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE5	DCMI_D6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE6	DCMI_D7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA4	DCMI_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA6	DCMI_PIXCLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC6	DCMI_D0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	DCMI_D1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD3	DCMI_D5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	DCMI_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE0	DCMI_D2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE1	DCMI_D3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
DEBUG	PA13	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	T_SWCLK
	PA15	DEBUG_JTDI	n/a	n/a	n/a	T_JTDI
	PB3	DEBUG_JTDO-SWO	n/a	n/a	n/a	T_SWO
	PB4	DEBUG_NJTRS T	n/a	n/a	n/a	T_NRST
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	DCMI_SCL
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	DCMI_SDA
I2S2	PC2_C	I2S2_SDI	Alternate Function Push Pull	No pull-up and no pull-down	Low	MIC_SDI
	PA11	I2S2_WS	Alternate Function Push Pull	No pull-up and no pull-down	Low	MIC_WS
	PA12	I2S2_CK	Alternate Function Push Pull	No pull-up and no pull-down	Low	MIC_CK
OCTOSPI1	PC3_C	OCTOSPIM_P1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PA1	OCTOSPIM_P1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PA3	OCTOSPIM_P1_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PA7	OCTOSPIM_P1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PB0	OCTOSPIM_P1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PB10	OCTOSPIM_P1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC1	PC8	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
SPI4	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	DCMI_SCK
	PE11	SPI4_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Low	DCMI_NSS
	PE13	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	DCMI_MISO
	PE14	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	DCMI_MOSI
UART4	PA0	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	T_TX
	PD0	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	T_RX
UART5	PB12	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_S_RX
	PB13	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_S_TX
UART7	PE7	UART7_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_RXD
	PE8	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_TXD
	PE9	UART7_RTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_RTS
	PE10	UART7_CTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	RF_CTS
GPIO	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_10
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_09
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_08
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_06
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_04
	PD13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_03
	PD14	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	RF_GPIO_02
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RF_GPIO_01
	PA8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RGB_LED_R
	PA9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RGB_LED_G
	PA10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RGB_LED_B
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DCMI_RST
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DCMI_FSIN
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DCMI_PWDN

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA1 configuration

nothing configured in DMA service

4.4. BDMA2 configuration

nothing configured in DMA service

4.5. MDMA configuration

nothing configured in DMA service

4.6. NVIC configuration

4.6.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD and PVM interrupts through EXTI line	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
SPI2 global interrupt	unused		
SDMMC1 global interrupt	unused		
UART4 global interrupt	unused		
UART5 global interrupt	unused		
DCMI and PSSI global interrupt	unused		
FPU global interrupt	unused		
UART7 global interrupt	unused		
SPI4 global interrupt	unused		
OCTOSPI1 global interrupt	unused		
JPEG global interrupt	unused		
HSEM1 global interrupt	unused		
Clock Recovery System global interrupt	unused		

4.6.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Pendable request for system service	false	true	false
System tick timer	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware									
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA1			DCI ✓	DCMI ✓			DEBUG ✓		
BDMA2			OCTOSPI1 ✓	DS2 ✓					
CORTEX_M7 ✓			SDMMC1 ✓	JPEG ✓					
DMA			SPW ✓						
GPIO ✓			UART4 ✓						
MDMA			UART5 ✓						
NVIC ✓			UART7 ✓						
ROC ✓									
SYS ✓									

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7a3-b3_line_product-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Advisory	https://www.st.com/resource/en/security_advisory/sa0045-security-firmware-install-no-longer-supported-on-stm32h7a3xx-and-stm32h7bxxx-microcontrollers-stmicroelectronics.pdf

Security Advisory	https://www.st.com/resource/en/security_advisory/sa0053-arbitrary-code-execution-with-privilege-on-stm32h73xxx-and-stm32h7bxxx-microcontrollers-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5307-getting-started-

with-stm32h7a37b3-line-and-stm32h7b0-value-line-microcontroller-hardware-development-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5450-stm32h7a37b3-lines-and-stm32h7b0-value-line-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

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- Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
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- Application Notes https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
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- Application Notes https://www.st.com/resource/en/application_note/an4943-how-to-use-chromart-accelerator-to-refresh-an-lcdtft-display-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
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Application Notes https://www.st.com/resource/en/application_note/an3155-how-to-use-uart-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf

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Application Notes	https://www.st.com/resource/en/application_note/an4776-how-to-use-generalpurpose-timer-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-introduction-to-debug-toolbox-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5050-getting-started-with-octospi-hexadecapi-and-xspi-interfaces-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5601-introduction-to-hdp-secure-area-for-stm32h7b3-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-how-to-use-i3c-protocol-in-bootloader-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-16bit-adc-on-stm32h7-mcus-stmicroelectronics.pdf
Design Notes & Tips	https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0478-stm32h7a3xig-stm32h7b0xb-and-stm32h7b3xi-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00661941.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0455-stm32h7a37b3-and-stm32h7b0-value-line-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf

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Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-devices-in-bga-packages-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2840-stm32h7-dualcore-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2331-stm32h7-singlecore-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3252-stm32h7-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf